

SoulScope Constellation State Registry v0.1

Status: Canonical scientific implementation companion — architecture final, state-region calibration pending

Governing authority: SoulScope Canon v1.3

Upstream: Constellation Dimension Registry v0.1; Inference Rule Registry v0.1

Downstream: Interaction Registry; Pattern Registry; Narrative Registry

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NORTH STAR

States summarize recurring geometry. They do not replace it.

SoulScope SHALL maintain a compact grammar of useful, evidence-supported state regions.

It SHALL NOT attempt to assign a unique named state to every possible four-point combination.

1. Purpose

The State Registry defines how completed Constellation Dimension posteriors become:

1. candidate canonical states;
2. state fit estimates;
3. boundary blends;
4. unresolved state outputs;
5. optional state qualifiers;
6. a traceable state decision.

The governing sequence is:

Dimension Posteriors → Continuous Constellation Geometry → Candidate State Evaluation → State / Blend / U

State reasoning occurs only after Dimension inference is complete.

2. The anti-combinatorial rule

For a four-point Constellation, even a crude five-band discretization would produce:

$$5^4 = 625$$

possible combinations.

Across four Constellations, manually naming every combination would create thousands of potential semantic cases before uncertainty, missingness, boundary overlap, and temporal movement were considered.

SoulScope explicitly rejects this architecture.

Instead:

Continuous Geometry + Compact Anchor State + Optional Qualifiers

produces semantic richness without inventing a state for every coordinate combination.

3. State hierarchy

A completed Constellation result contains four distinct layers.

Layer 1 — Continuous Geometry

The authoritative four-point posterior structure.

Layer 2 — Canonical Anchor State

A compact reference region representing a recurring, useful configuration.

Layer 3 — Qualifiers

Secondary resolved Dimensions that add useful information without creating a new state identity.

Layer 4 — Interaction

Relationships with other independently resolved Constellations.

Therefore:

Anchor State + Qualifiers is preferred over state-name proliferation.

4. State object

Every registered state SHALL contain:

```
state_id
state_version

constellation_id

canonical_internal_name
scientific_status

required_dimensions[]
optional_dimensions[]
excluded_conditions[]

region_model_type
region_model_version

adjacent_states[]

permitted_qualifiers[]

minimum_dimension_coverage
minimum_state_fit
minimum_state_confidence

blend_policy

prohibited_meanings[]

activation_status
validation_status
```

5. State scientific status

Canonical statuses:

```
ENGINEERING_PRIOR
RESEARCH_ONLY
RELIABILITY_VERIFIED
CONSTRUCT_VALIDATED
CALIBRATED
PRODUCTION_ACTIVE
DEPRECATED
```

A State may not have a stronger scientific status than the weakest required Dimension on which it depends.

Formally:

$$Status(S) \leq \min_{d \in R_S} Status(d)$$

where R_S is the required Dimension set.

6. Anchor States versus Qualifiers

The State Engine SHALL distinguish:

state identity

from:

secondary characteristics.

Example:

Anchor State:
COG-S01 CONTINUOUS_ORGANIZATION

Qualifiers:
HIGHER_PROCESSING_DEMAND
BROADER_EXPLORATION

SoulScope SHALL NOT create separate states such as:

Continuous Organization With High Demand
Continuous Organization With High Demand And Exploration
Continuous Organization With Exploration And Moderate Demand

merely because those combinations occur.

The secondary Dimensions remain qualifiers.

7. Continuous geometry remains authoritative

For Constellation C :

$$C = (D_1, D_2, D_3, D_4)$$

where each Dimension is itself:

$$D_i = \{\mu_i, L_i, U_i, C_i, E_i, B_i, K_i, H_i, M_i, R_i\}.$$

State selection does not replace this object.

The full Dimension geometry SHALL remain stored even after a State is selected.

8. Missing Dimensions

Missing or unresolved Dimensions SHALL remain unresolved.

A state may be evaluated only when all Dimensions listed as:

required_dimensions

are resolved and publication-eligible.

Optional unresolved Dimensions:

- do not become 0.5;
 - do not become neutral;
 - do not become low;
 - do not automatically disqualify the State;
 - cannot generate qualifiers.
-

9. Current protocol consequence

Under the current three-prompt protocol:

REG-P4 Recovery
CAP-P2 Reserve
EXP-P4 Relational Availability

remain structurally unresolved.

Therefore current-protocol canonical States SHALL NOT require those Dimensions.

Their absence SHALL NOT prevent valid State reasoning based on other authorized Dimensions.

Their missing values also SHALL NOT be inferred from the selected State.

10. State-region principle

Canonical States are **compact reference regions**.

They are not exhaustive partitions.

The geometry outside a registered State region is allowed to remain:

BOUNDARY_BLEND

or:

UNRESOLVED

There is no requirement that every mathematically possible point belong to one State.

11. State region representation

The registry SHALL support calibrated multivariate region models.

Preferred production representation:

ELLIPSOIDAL_PROTOTYPE

with:

$$\mu_S$$

as the State prototype and:

$$\Sigma_S$$

as its calibrated covariance or distance structure over required Dimensions.

Other permitted types include:

RULE_REGION
CALIBRATED_DENSITY
MIXTURE_REGION

provided the model remains versioned and explainable.

12. Posterior-aware State fit

State matching SHALL use Dimension uncertainty, not only Dimension means.

For State S and Dimension posterior $p_d(x)$, define State membership function:

$$\phi_{S,d}(x) \in [0, 1].$$

Dimension-level overlap is:

$$O_{S,d} = \int_0^1 p_d(x) \phi_{S,d}(x) dx.$$

This permits:

- narrow high-confidence posterior strongly inside region → strong overlap;
 - broad uncertain posterior spanning boundary → weaker overlap;
 - posterior outside region → low overlap.
-

13. Multidimensional State fit

For required Dimensions:

$$Fit_S = \exp \left(\frac{\sum_d w_d \log(\max(O_{S,d}, \epsilon))}{\sum_d w_d} \right).$$

This geometric aggregation is preferred conceptually because one severely incompatible required Dimension should materially reduce State fit.

Exact aggregation remains calibration-dependent.

14. State fit is not automatically probability

Until empirically calibrated:

state_fit

SHALL be described as:

compatibility with the registered State region.

It SHALL NOT automatically be displayed or interpreted as:

probability that the person is in this state.

Only a formally calibrated probabilistic model may use probability terminology.

15. State coverage

For State S :

$$Coverage_S = \frac{\sum_{d \in R_S} w_d I_d}{\sum_{d \in R_S} w_d}.$$

Every required Dimension must satisfy the active coverage policy.

A missing required Dimension cannot be compensated for by stronger fit on another Dimension.

16. State confidence

State confidence is separate from State fit.

Conceptually:

$$C_S = f(Fit_S, C_D, Coverage_S, Separation_S, K_S, Validation_S)$$

where:

- C_D = underlying Dimension confidence;
- $Separation_S$ = distance from plausible alternative States;
- K_S = contradiction burden;
- $Validation_S$ = State-model calibration quality.

A State may have:


```
fit = high
confidence = moderate
```

when the geometry points toward the region but uncertainty remains large.

17. Candidate State eligibility

A State becomes a candidate only if:

1. the State is active under the current protocol;
 2. all required Dimensions are resolved;
 3. all required Dimensions meet publication status;
 4. State model version is compatible;
 5. no exclusion rule fires;
 6. required fit threshold is met;
 7. required State confidence threshold is met;
 8. no blocking confound exists.
-

18. Independent candidate evaluation

State fits SHALL NOT be normalized in a way that forces:

$$\sum_S P(S) = 1$$

unless a validated probabilistic State model specifically justifies that assumption.

Each State is evaluated independently.

Therefore all candidate States may legitimately have low fit.

When that occurs:

UNRESOLVED

is the correct answer.

19. Candidate rejection

For every rejected State, the Decision Ledger SHALL preserve:

```
state_id

fit

requirements_passed[]
requirements_failed[]

supporting_dimensions[]
contradicting_dimensions[]
missing_dimensions[]

exclusion_rules_triggered[]

rejection_reason
```

State selection must be reconstructable.

20. Boundary blend

When two adjacent canonical State regions are both well supported and neither clearly dominates, SoulScope SHALL preserve the overlap.

Canonical result:

```
BOUNDARY_BLEND
```

rather than forcing a single winner.

21. Blend eligibility

A two-State blend requires:

1. both States are eligible;
 2. both exceed the active blend-fit minimum;
 3. the States are declared adjacent;
 4. fit separation is within the calibrated blend margin;
 5. neither State has a decisive exclusion;
 6. underlying geometry genuinely overlaps both regions.
-

22. Adjacency must be explicit

A State may blend only with States listed in:

```
adjacent_states[]
```

This prevents arbitrary semantic mixtures.

If two strongly plausible States are **not** adjacent:

```
UNRESOLVED
```

is preferred.

That means the geometry is not adequately represented by the current grammar.

23. More than two plausible States

If more than two State regions remain materially plausible:

```
UNRESOLVED_MULTIPLE_CANDIDATES
```

is preferred unless a separately validated multi-region boundary rule exists.

SoulScope SHALL NOT create a three-name compound State merely to avoid abstention.

24. Qualifier architecture

A qualifier is a secondary statement derived from a resolved Dimension.

Canonical qualifier object:

```
qualifier_id  
qualifier_version  
  
dimension_id  
constellation_id
```

```
direction_or_region

minimum_dimension_confidence
minimum_posterior_mass

compatible_states[]
incompatible_states[]

user_language_key

scientific_status
```

25. Qualifier rule

Qualifiers SHALL be:

- independently supported;
- secondary to State identity;
- omitted when uncertain;
- limited in number in user-facing presentation.

Recommended user-facing maximum:

2

qualifiers per Constellation summary.

Additional resolved information remains available through progressive disclosure.

26. Qualifiers do not alter State identity

Example:

```
State:
  REG-S01 STABLE_PATTERN

Qualifier:
  HIGHER_ACTIVATION
```

does not create a new registered State:

STABLE_HIGH_ACTIVATION_PATTERN

unless future evidence demonstrates that configuration is a genuinely distinct, reproducible, scientifically useful reference region.

27. State promotion criterion

A new canonical State should be added only when data demonstrate that the candidate region is:

1. recurrent;
2. statistically distinguishable from existing States;
3. sufficiently reliable;
4. semantically useful;
5. not adequately expressible as an existing State plus qualifiers;
6. evidence-supported across relevant populations;
7. understandable without encouraging overinterpretation;
8. worth the maintenance and validation cost of another State.

HARD RULE

A configuration does not deserve a name merely because it exists geometrically.

28. State deletion criterion

A State should be deprecated when:

- it cannot be reliably distinguished;
- it rarely occurs outside another State's boundary;
- it is better represented by a qualifier;
- it produces systematic user misunderstanding;
- scientific validation fails;
- it no longer corresponds to the active Dimension architecture.

Historical results retain the deprecated State/version.

29. Internal versus user-facing names

Every State has:

canonical_internal_name

separate from:

user_language_key

Internal names SHALL be scientific/descriptive.

User-facing wording belongs to the Narrative Registry.

A narrative-copy revision SHALL NOT silently change State geometry.

30. Prohibited State meanings

No State may claim:

- diagnosis;
- personality type;
- emotional identity;
- attachment style;
- trauma status;
- deception;
- honesty;
- mental-health condition;
- cognitive ability;
- moral characteristic;
- spiritual status;
- permanent personal identity.

States describe the **current scan geometry**.

31. Cognitive Form State Grammar

Current Dimensions:

COG-P1 Organization
COG-P2 Exploration
COG-P3 Focus Continuity
COG-P4 Processing Demand

The compact grammar prioritizes Organization and Focus Continuity as core structural dimensions.

Exploration and Processing Demand primarily function as qualifiers unless future validation demonstrates a distinct recurring State region.

32. COG-S01 – CONTINUOUS_ORGANIZATION

Status: Initial canonical anchor

Current scientific status: Engineering prior pending calibration

Required Dimensions

COG-P1 Organization
COG-P3 Focus Continuity

Prototype meaning

A geometry characterized by comparatively stronger:

- observable speech-production organization;
- continuity of ongoing speech production.

Optional qualifiers

COG-Q01 BROADER_EXPLORATION
COG-Q02 NARROWER_EXPLORATION

COG-Q03 HIGHER_PROCESSING_DEMAND
COG-Q04 LOWER_PROCESSING_DEMAND

Example structural outputs

CONTINUOUS_ORGANIZATION
+ HIGHER_PROCESSING_DEMAND

or:

CONTINUOUS_ORGANIZATION
+ BROADER_EXPLORATION

The system SHALL NOT create a separate State ID for either combination.

33. COG-S02 – FRAGMENTED_ORGANIZATION

Status: Initial canonical anchor

Current scientific status: Engineering prior pending calibration

Required Dimensions

COG-P1 Organization
COG-P3 Focus Continuity

Prototype meaning

A geometry characterized by comparatively lower observable:

- speech-production organization;
- continuity of speech output.

Important boundary

The name describes vocal-output geometry.

It does not mean:

disorganized thinking
thought disorder
poor concentration
cognitive impairment

Optional qualifiers

Same COG qualifier set.

34. COG-S03 – EXPLORATORY_REORGANIZATION

Status: Research candidate State

Required Dimensions

COG-P1 Organization
COG-P2 Exploration
COG-P3 Focus Continuity

Candidate prototype

A recurring configuration in which:

- Exploration is comparatively elevated;
- speech remains sufficiently continuous;
- organizational structure shifts without becoming strongly fragmented.

Activation rule

This State SHALL remain:

RESEARCH_ONLY

until COG-P2 Exploration is construct-validated and State clustering demonstrates that this region provides information not adequately expressed as a qualifier.

35. COG State adjacency

Initial adjacency graph:

COG-S01 <-> COG-S02

COG-S03 <-> COG-S01
COG-S03 <-> COG-S02

COG-S03 remains research-only until activated.

36. Regulatory Motion State Grammar

Current Dimensions:

REG-P1 Activation
REG-P2 Stability
REG-P3 Flexibility
REG-P4 Recovery

Under the current protocol:

REG-P4 Recovery = UNRESOLVED

and cannot contribute to State identity.

Stability provides the primary anchor geometry.

Activation and Flexibility provide secondary structure.

37. REG-S01 – STABLE_PATTERN

Status: Initial canonical anchor

Required Dimension

REG-P2 Stability

Prototype meaning

Comparatively stable vocal organization within the observed conditions.

Optional qualifiers

REG-Q01 HIGHER_ACTIVATION
REG-Q02 LOWER_ACTIVATION

REG-Q03 GREATER_CONTEXTUAL_FLEXIBILITY
REG-Q04 LOWER_CONTEXTUAL_FLEXIBILITY

Possible output:

```
STABLE_PATTERN
+ HIGHER_ACTIVATION
+ GREATER_CONTEXTUAL_FLEXIBILITY
```

This does not become a new State.

38. REG-S02 – VARIABLE_PATTERN

Status: Initial canonical anchor

Required Dimension

```
REG-P2 Stability
```

Prototype meaning

Comparatively greater short-timescale variability or volatility in qualified vocal organization.

Prohibited meaning

It does not mean:

```
emotionally unstable
psychologically unstable
dysregulated
mentally unwell
```

Optional qualifiers

Same REG qualifier set.

39. REG-S03 – FLEXIBLE_RECONFIGURATION

Status: Research candidate State

Required Dimensions

REG-P2 Stability
REG-P3 Flexibility

Candidate prototype

Strong contextual reconfiguration combined with sufficient within-condition coherence.

Conceptually:

Flexibility / random variability.

The region should represent:

context-sensitive change + coherence.

Activation rule

Requires validated REG-P3 construct and evidence that the region adds meaning beyond:

STABLE_PATTERN + GREATER_CONTEXTUAL_FLEXIBILITY

Until then:

RESEARCH_ONLY

40. REG-S04 – CONSTRAINED_RECONFIGURATION

Status: Research candidate State

Required Dimensions

REG-P2 Stability
REG-P3 Flexibility

Candidate prototype

Comparatively stable vocal organization accompanied by limited contextual reconfiguration.

Boundary

This does not mean:

rigid personality
poor coping
emotional suppression

Activation rule

Same as REG-S03.

41. REG State adjacency

Initial graph:

REG-S01 <-> REG-S02

REG-S03 <-> REG-S01
REG-S03 <-> REG-S02

REG-S04 <-> REG-S01

Research States do not enter production blends until production-active.

42. Available Capacity State Grammar

Current Dimensions:

CAP-P1 Mobilization
CAP-P2 Reserve
CAP-P3 Effort Cost
CAP-P4 Sustainability

Current protocol:

CAP-P2 Reserve = UNRESOLVED

Reserve SHALL NOT contribute to current State selection.

Sustainability is the primary anchor.

Mobilization and Effort Cost provide qualifiers.

43. CAP-S01 – MAINTAINED_OUTPUT

Status: Initial canonical anchor

Required Dimension

CAP-P4 Sustainability

Prototype meaning

Observable vocal organization remained comparatively maintained across the limited interval measured.

Optional qualifiers

CAP-Q01 HIGHER_MOBILIZATION
CAP-Q02 LOWER_MOBILIZATION

CAP-Q03 HIGHER_EFFORT_COST
CAP-Q04 LOWER_EFFORT_COST

Possible output:

MAINTAINED_OUTPUT
+ HIGHER_MOBILIZATION
+ HIGHER_EFFORT_COST

This is intentionally different from inventing:

COSTLY_MAINTAINED_MOBILIZATION

as a new State.

44. CAP-S02 – REDUCED_MAINTENANCE

Status: Initial canonical anchor

Required Dimension

CAP-P4 Sustainability

Prototype meaning

Observable vocal organization showed comparatively less maintenance across the limited interval measured.

Mandatory scope language

Any narrative derived from this State SHALL make clear that the observation applies to:

the recorded response interval.

It does not describe general endurance or chronic capacity.

45. CAP-S03 – COSTLY_MAINTENANCE

Status: Research candidate State

Required Dimensions

CAP-P3 Effort Cost
CAP-P4 Sustainability

Candidate prototype

Maintenance of observable output accompanied by comparatively elevated acoustic/timing cost.

Activation rule

The State may become canonical only if validation demonstrates that:

```
MAINTAINED_OUTPUT + HIGHER_EFFORT_COST
```

fails to adequately represent the recurring region.

Until then:

```
RESEARCH_ONLY
```

46. CAP State adjacency

Initial graph:

```
CAP-S01 <-> CAP-S02  
  
CAP-S03 <-> CAP-S01  
CAP-S03 <-> CAP-S02
```

47. Expressive Interface State Grammar

Current Dimensions:

```
EXP-P1 Range  
EXP-P2 Openness  
EXP-P3 Restraint  
EXP-P4 Relational Availability
```

Current protocol:

```
EXP-P4 Relational Availability = UNRESOLVED
```

Range is the primary anchor.

Openness and Restraint provide contextual deployment information.

48. EXP-S01 — BROAD_MODULATION

Status: Initial canonical anchor

Required Dimension

EXP-P1 Range

Prototype meaning

Comparatively broad qualified acoustic modulation was observed during the scan.

Optional qualifiers

EXP-Q01 GREATER_DEPLOYMENT

EXP-Q02 LOWER_DEPLOYMENT

EXP-Q03 GREATER_CONTEXTUAL_RESTRAINT

EXP-Q04 LOWER_CONTEXTUAL_RESTRAINT

49. EXP-S02 — NARROW_MODULATION

Status: Initial canonical anchor

Required Dimension

EXP-P1 Range

Prototype meaning

Comparatively narrower qualified acoustic modulation was observed.

Critical rule

Narrow Range / Restraint. =

Narrow modulation alone cannot generate a Restraint qualifier.

50. EXP-S03 – CONTEXTUAL_DEPLOYMENT

Status: Research candidate State

Required Dimensions

EXP-P1 Range
EXP-P2 Openness

Candidate prototype

Available modulation is clearly deployed across relevant contexts.

Activation requirement

SoulScope must demonstrate that this configuration represents a reproducible region beyond:

BROAD_MODULATION + GREATER_DEPLOYMENT

Until then:

RESEARCH_ONLY

51. EXP-S04 – CONTEXTUAL_COMPRESSION

Status: Research candidate State

Required Dimensions

EXP-P1 Range
EXP-P3 Restraint

Candidate prototype

Broader modulation is demonstrably available, while a compatible context shows relative compression.

Hard prerequisite

Evidence of available range must exist.

If broader available range is not established:

```
EXP-P3 Restraint = UNRESOLVED
```

and this State is ineligible.

52. EXP State adjacency

Initial graph:

```
EXP-S01 <-> EXP-S02  
  
EXP-S03 <-> EXP-S01  
EXP-S03 <-> EXP-S02  
  
EXP-S04 <-> EXP-S01  
EXP-S04 <-> EXP-S02
```

Research States remain excluded from production candidate sets until activated.

53. Current canonical anchor set

The intentionally compact initial grammar therefore contains eight primary production-oriented anchors:

```
COG-S01 CONTINUOUS_ORGANIZATION  
COG-S02 FRAGMENTED_ORGANIZATION  
  
REG-S01 STABLE_PATTERN  
REG-S02 VARIABLE_PATTERN  
  
CAP-S01 MAINTAINED_OUTPUT
```

CAP-S02 REDUCED_MAINTENANCE

EXP-S01 BROAD_MODULATION

EXP-S02 NARROW_MODULATION

and seven higher-order research candidates:

COG-S03 EXPLORATORY_REORGANIZATION

REG-S03 FLEXIBLE_RECONFIGURATION

REG-S04 CONSTRAINED_RECONFIGURATION

CAP-S03 COSTLY_MAINTENANCE

EXP-S03 CONTEXTUAL_DEPLOYMENT

EXP-S04 CONTEXTUAL_COMPRESSION

The small initial grammar is intentional.

54. Why eight primary anchors are enough initially

The semantic richness comes from:

Anchor + Qualifiers + Geometry + Momentum + Cross-Constellation Interactions.

For example:

COG

CONTINUOUS_ORGANIZATION

+ HIGHER_PROCESSING_DEMAND

REG

STABLE_PATTERN

+ HIGHER_ACTIVATION

+ GREATER_CONTEXTUAL_FLEXIBILITY

CAP

MAINTAINED_OUTPUT

+ HIGHER_EFFORT_COST

```
EXP  
BROAD_MODULATION  
+ GREATER_CONTEXTUAL_RESTRAINT
```

Already describes a highly specific scan without creating four bespoke State names.

55. State qualifier result

Canonical representation:

```
qualifiers:  
- qualifier_id  
  source_dimension_id  
  direction  
  posterior_support  
  confidence  
  evidence_trace
```

Qualifiers remain traceable to their Dimension Decision Records.

56. Qualifier abstention

A qualifier SHALL be omitted when:

- source Dimension is unresolved;
- confidence is insufficient;
- posterior overlaps qualifier boundaries too broadly;
- qualifier construct is not production-active;
- a required prerequisite is missing.

An omitted qualifier does not imply its opposite.

57. State Decision Record

Every Constellation creates:

```
ConstellationStateDecision
```

```
constellation_id

geometry_snapshot

eligible_states[]
rejected_states[]

candidate_fit[]

candidate_confidence[]

top_candidate
second_candidate

adjacency_status

blend_evaluation

selected_state
selected_blend

qualifiers[]

publication_status

primary_reason
secondary_reasons[]

state_registry_version
region_model_version
```

58. State outcome types

Canonical outcomes:

```
STATE
BOUNDARY_BLEND
UNRESOLVED
SUPPRESSED
INVALID
```

59. STATE outcome

Returned when one eligible canonical State:

- exceeds publication requirements;
 - is sufficiently separated from alternatives;
 - has adequate State confidence;
 - has no blocking contradiction.
-

60. BOUNDARY_BLEND outcome

Example:

```
type = BOUNDARY_BLEND  
  
state_a = COG-S01  
state_b = COG-S02  
  
weight_a = calibrated  
weight_b = calibrated
```

Blend weights SHALL NOT automatically be interpreted as State probabilities.

The user narrative should communicate overlap rather than numerical precision unless that precision has been validated.

61. UNRESOLVED outcome

Canonical reasons include:

```
NO_STATE_EXCEEDS_FIT_THRESHOLD  
LOW_STATE_CONFIDENCE  
MISSING_REQUIRED_DIMENSION  
MULTIPLE_NONADJACENT_CANDIDATES  
MORE_THAN_TWO_PLAUSIBLE_STATES  
HIGH_DIMENSION_CONTRADICTION  
STATE_MODEL_NOT_VALIDATED  
STATE_MODEL_OUT_OF_DISTRIBUTION
```

62. Unresolved geometry remains useful

An unresolved State does not erase the Constellation.

SoulScope may still retain and, where appropriate, explain:

- resolved Dimensions;
- uncertainty;
- movement;
- Evidence;
- comparisons.

State abstention applies only to the State label.

63. State temporal movement

The system SHALL preserve movement of continuous geometry independently from State identity.

For scans $t_1, t_2, \dots, t_n$:

$$C(t) = (D_1(t), D_2(t), D_3(t), D_4(t)).$$

Longitudinal analysis may describe:

```
MOVING_TOWARD_STATE  
MOVING_AWAY_FROM_STATE  
CROSSING_BOUNDARY  
REMAINING_WITHIN_REGION  
STATE_TRANSITION  
INSUFFICIENT_COMPARABILITY
```

only where historical compatibility permits.

64. State transition is not improvement

No State is canonically defined as:

```
better  
worse  
healthier
```


unhealthier
more evolved
less evolved

Therefore:

$$S_A \rightarrow S_B$$

means the measured geometry changed.

It does not by itself mean improvement or deterioration.

65. State desirability

Canonical States SHALL NOT have intrinsic positive or negative valence.

For example:

STABLE_PATTERN

is not automatically desirable.

High stability may coexist with limited Flexibility.

Likewise:

VARIABLE_PATTERN

is not inherently undesirable.

Variation may be contextually appropriate.

Cross-Dimension and cross-Constellation relationships provide the richer interpretation.

66. Candidate-region calibration

State prototypes SHALL ultimately be estimated from SoulScope data.

Candidate approaches include:

- preregistered rule regions;
- hierarchical mixture models;
- latent-profile analysis;
- robust clustering followed by confirmatory validation;
- calibrated multidimensional density estimation.

Discovery does not equal validation.

A cluster found in one dataset does not automatically become a canonical State.

67. Discovery-to-State promotion

Candidate State discovery should follow:

Discovery Sample → Candidate Region → Independent Replication → Reliability Testing → Interpretability Review

68. Incremental-value test

Before adding State S_{new} , ask:

Does S_{new}

provide reproducible information that cannot be adequately represented by:

$S_{existing} + Qualifiers?$

If no:

DO_NOT_CREATE_STATE

69. State naming review

Before activation, every State name SHALL undergo:

- scientific-boundary review;
- user-understanding review;
- identity/trait-risk review;
- clinical-overreading review;

- translation/localization review.

A scientifically valid region may still require a different user-facing name.

70. State model versioning

A State requires a MAJOR version change when:

- required Dimensions change;
- State prototype meaning changes;
- region geometry materially changes;
- exclusion logic changes;
- State adjacency changes materially;
- scientific interpretation changes.

Minor numeric recalibration may follow the active semantic-version policy where historical compatibility is maintained.

71. Historical State results

Historical State decisions SHALL preserve:

```
state_id
state_version
region_model_version
dimension_versions
decision_version
```

A later State model SHALL NOT silently rewrite a historical State.

72. State reprocessing

If product policy permits recalculation using a newer State model, the result SHALL be represented as a separate interpretation:

```
original_result
reprocessed_result
```

with both versions identifiable.

73. State-engine hard invariants

STATE-INV-01

A State never creates Dimension values.

STATE-INV-02

Missing Dimensions are never imputed to satisfy a State.

STATE-INV-03

Every State fit is evaluated against the full posterior information available for required Dimensions.

STATE-INV-04

No candidate normalization may force a winner.

STATE-INV-05

Only declared adjacent States may blend.

STATE-INV-06

More than two materially plausible States normally produces unresolved.

STATE-INV-07

Qualifiers do not create new State identities.

STATE-INV-08

A new State requires demonstrated incremental value over State + qualifier grammar.

STATE-INV-09

State names do not imply diagnosis, identity, personality, or emotional truth.

STATE-INV-10

State scientific status cannot exceed the weakest required Dimension.

STATE-INV-11

D3 Dimensions cannot be inferred through State geometry.

STATE-INV-12

State output remains subordinate to continuous geometry.

74. Acceptance tests

STATE-MISS-001

Given:

```
required_dimension = UNRESOLVED
```

Expected:

```
candidate_state = INELIGIBLE
```

No imputation.

STATE-OPT-001

Given:

```
optional_dimension = UNRESOLVED
```

Expected:

- State may still be evaluated;
 - no qualifier from that Dimension.
-

STATE-FORCE-001

Given all candidate fits below threshold:

Expected:

UNRESOLVED

not highest candidate.

STATE-BLEND-001

Given two adjacent candidates with overlapping support inside blend margin:

Expected:

BOUNDARY_BLEND

STATE-BLEND-002

Given two nonadjacent candidates with similar high fit:

Expected:

UNRESOLVED

STATE-MULTI-001

Given three materially plausible candidates:

Expected:

UNRESOLVED_MULTIPLE_CANDIDATES

unless a separately validated policy exists.

STATE-D3-001

Given current protocol and a State requiring Recovery:

Expected:

STATE_INELIGIBLE

STATE-QUAL-001

Given:

CONTINUOUS_ORGANIZATION
Processing Demand = higher, resolved

Expected:

State = CONTINUOUS_ORGANIZATION
Qualifier = HIGHER_PROCESSING_DEMAND

not creation of a new State ID.

STATE-QUAL-002

Given:

Processing Demand = UNRESOLVED

Expected:

no Processing Demand qualifier

not LOWER_PROCESSING_DEMAND .

STATE-EXP-001

Given:

EXP Range = narrow
Restraint = unresolved

Expected:

```
State = NARROW_MODULATION
```

with no Restraint inference.

STATE-CAP-001

Given:

```
CAP Sustainability = high  
Effort Cost = high
```

Expected initial grammar:

```
State = MAINTAINED_OUTPUT  
Qualifier = HIGHER_EFFORT_COST
```

until `CAP-S03 COSTLY_MAINTENANCE` independently earns canonical activation.

75. Example complete COG result

Constellation:
COG

Dimensions:
Organization:
mean: 0.74
interval: [0.66, 0.82]
confidence: high

Exploration:
mean: 0.63
interval: [0.47, 0.76]
confidence: moderate
status: research-only

Focus Continuity:
mean: 0.79

interval: [0.71, 0.86]
confidence: high

Processing Demand:
mean: 0.68
interval: [0.56, 0.77]
confidence: moderate
status: production-active after validation

Candidate States:
CONTINUOUS_ORGANIZATION:
fit: strong

FRAGMENTED_ORGANIZATION:
fit: low

Result:
CONTINUOUS_ORGANIZATION

Qualifiers:
HIGHER_PROCESSING_DEMAND

Exploration:
retained in geometry
not published as qualifier if not production-active

76. Example complete REG result

Stability:
higher
high confidence

Activation:
higher
moderate-high confidence

Flexibility:
higher
moderate confidence

Recovery:
unresolved
NO_RECOVERY_COMPATIBLE_CONDITION

Result:

```
STABLE_PATTERN
+ HIGHER_ACTIVATION
+ GREATER_CONTEXTUAL_FLEXIBILITY
```

Recovery remains unresolved.

The selected State SHALL NOT imply recovery.

77. Example boundary result

Suppose the posterior for COG Organization and Focus Continuity spans the transition between the two primary anchors.

Candidate results:

```
COG-S01 fit = 0.69
COG-S02 fit = 0.66
```

and the two States are adjacent.

If the active calibrated blend policy permits:

```
BOUNDARY_BLEND:
  CONTINUOUS_ORGANIZATION
  FRAGMENTED_ORGANIZATION
```

The narrative may say:

The pattern sat between more continuous organization and greater interruption rather than landing clearly in one region.

No artificial winner is required.

78. Example unresolved result

Suppose:

Organization:
broad interval

Focus Continuity:
contradictory Evidence

Exploration:
research-only

Processing Demand:
unresolved

Then:

COG State:
UNRESOLVED

Reason:
LOW_REQUIRED_DIMENSION_CONFIDENCE
+ HIGH_DIMENSION_CONTRADICTION

The underlying Dimension information remains available.

79. Relationship to the Interaction Engine

State selection finishes **within-Constellation reasoning**.

Only after each Constellation has produced:

State / Blend / Unresolved
+
Dimension geometry
+
Qualifiers

may the Interaction Engine evaluate relationships across Constellations.

The Interaction Engine SHALL receive the continuous Dimension structure in addition to State labels.

It SHALL NOT reason only from State names.

80. Relationship to the Pattern Engine

The Pattern Engine receives:

```
Constellation geometries
State results
Boundary blends
Qualifiers
Interactions
Confidence
Contradiction
Missingness
```

A Pattern is therefore downstream of complete reasoning.

It does not create State meaning.

81. Relationship to Narrative

The Narrative layer may translate:

```
Anchor State
+
Qualifier
+
Interaction
+
Evidence trace
```

into natural language.

It SHALL NOT invent a more specific State because the combination sounds narratively appealing.

82. Relationship to the Resonance Signature

There is none at the semantic geometry level.

The Resonance Signature does not consume State identity.

The State Engine does not consume Resonance Signature geometry.

Both remain linked to the same scan through the immutable result envelope.

83. Machine-readable State Registry

Recommended structure:

```
states:
  - state_id
  - state_version
  - constellation_id

  - required_dimensions
  - optional_dimensions

  - region_model
  - region_parameters

  - adjacency

  - permitted_qualifiers

  - hard_exclusions

  - publication_policy

  - scientific_status

qualifiers:
  - qualifier_id
  - source_dimension
  - region
  - confidence_policy
  - compatible_states

global_policy:
  - blend_policy
  - unresolved_policy
  - candidate_policy
  - version_policy
```

84. What v0.1 freezes

The following architecture is now canonical:

- continuous geometry is primary;
 - States are compact reference regions;
 - States do not exhaust the geometry;
 - State fit uses Dimension posterior information;
 - required and optional Dimensions are distinct;
 - missing required Dimensions block a candidate;
 - missing optional Dimensions remain missing;
 - candidates are independently evaluated;
 - no forced winner;
 - explicit adjacency;
 - two-State boundary blends;
 - multi-State ambiguity defaults to unresolved;
 - State + qualifiers replaces combinatorial State naming;
 - State status cannot exceed required Dimension status;
 - D3 Dimensions cannot be recovered through geometry;
 - new States require incremental value;
 - State decisions are fully traceable.
-

85. What v0.1 intentionally does not freeze

The following remain empirical:

- exact State centroids;
- covariance matrices;
- region radii;
- membership functions;
- State-fit thresholds;
- confidence thresholds;
- blend margins;
- qualifier posterior thresholds;
- final active research candidate States;
- user-facing State names.

These should be calibrated from SoulScope data rather than invented permanently in advance.

86. Initial production strategy

SoulScope should begin with the **smallest useful State grammar**.

The primary anchor set should be treated as a hypothesis to test, not a marketing requirement.

If validation demonstrates that:

State + Qualifiers

explains the geometry well, the grammar should remain small.

If repeated evidence demonstrates genuinely distinct regions, new States may be promoted carefully.

The success metric is not:

How many States does SoulScope have?

It is:

How much useful structure can SoulScope describe accurately without pretending the evidence is more categorical than it is?

87. Final State law

Geometry first

State second

Qualifier before new State

Blend before forced winner

Unresolved before invented precision

A State earns its name only when the underlying geometry repeatedly earns the distinction.